

CHAPTER 2

Immunity to the COVID-19 shock? The case of US innovation

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1 INTRODUCTION

The coronavirus pandemic has created much hardship in health and care for families and communities, strained hospital facilities, disrupted school systems, and restricted travel and other social activities. The economic consequences have also been dire. The temporary ‘lockdowns’ of several business operations, schools, and certain services that were put into place to help contain the virus led immediately to a sharp recession, with double-digit unemployment rates not seen in decades.¹ The economic effects still linger in early 2022, with inflationary pressures brought about by disruptions to supply chains, labour shortages, and many pandemic restrictions that are still in effect.²

Yet, amid all the economic upheaval, innovation in the US economy has shown remarkable resilience. Research and development (R&D) investments by companies, flows of inventions, and new company formations have been robust. In this chapter, we analyse the performance of US innovation during the COVID-19 era. We present data on R&D, patenting, and trademark applications and discuss the factors that helped to maintain the strength of US innovation during the pandemic.

Three takeaways about the COVID-19 experience and innovation in the US are:

- The output of goods and services in industries that use patents, trademarks, and copyrights intensively experienced less of a decline than in other industries.
- Investments in knowledge (e.g. R&D) showed greater stability than investments in physical capital as economic uncertainty increased due to the pandemic.
- The ability to obtain intellectual property protection allowed companies to store value and recoup investments in knowledge, but recoupment opportunities vary by technological field.

¹ Information on the dates that US states implemented shelter-in-place orders is available from the Kaiser Family Foundation at www.kff.org/policy-watch/stay-at-home-orders-to-fight-covid19/.

² The Census Bureau, through its Small Business Pulse Survey (SBPS), has been monitoring the economic impacts (Buffington et al. 2021a). The survey makes clear the *heterogeneity* across industry groups. For example, construction and finance and insurance have been able to return to normal operations much sooner than manufacturing and accommodation and food services.

US companies that invest in knowledge-generating activities, rely more heavily on high-skilled labour and use relatively more intellectual property weathered the COVID-19 recession substantially better. Part of the reason is that lockdown policies differentially hurt companies that rely on face-to-face business relationships by suddenly and dramatically reducing the demand for their products and services. Other policies, such as Operation Warp Speed, the Coronavirus Aid, Relief, and Economic Security (CARES) Act, and the Paycheck Protection Program, provided trillions in loans and grants to stimulate vaccine research, help American citizens, and prop up US businesses.³ However, innovative companies enjoyed another ‘shot of immunity’ stemming from the nature of knowledge investments and intellectual property. Investments in knowledge such as R&D are often directed at longer-term competitive objectives as opposed to meeting immediate customer needs. The information gleaned from these investments tends to be more adaptable than physical structures and equipment that result from tangible capital investments. The results from investments in knowledge can often be repurposed and recovered, and are therefore not necessarily lost or sunk costs. Moreover, intellectual property can be used to store value during times of uncertainty. Patents, trademarks, and copyrights are tradeable assets that can be held to be sold in the future when economic circumstances improve.

To investigate the overall performance of US companies that innovate, we examine real gross output (in 2012 dollars) produced by companies in intellectual property (IP)-intensive industries against that of non-IP-intensive industries during the time of COVID-19 (Figure 1).⁴ We normalise real gross output to the fourth quarter of 2019, giving that quarter an indexed value of one for IP-intensive industries, non-IP-intensive, and the overall economy (i.e. the combined IP-intensive and non-IP-intensive industries). This quarter was chosen because the first cases of infection in the US occurred in January of 2020 and segments of the country started going into lockdown in March of 2020.⁵

Figure 1 shows that companies in IP-intensive industries suffered a smaller output loss following the initial COVID-19 shock than did the non-IP-intensive industries. Signs of economic trouble emerged around the end of the first quarter of 2020, so that by the second quarter of 2020, the US economy experienced recessionary pressures from business shutdowns, job losses, and consumer pessimism.⁶ In the second quarter of 2020, output in IP-intensive industries was a little over 90% of what it was in the last quarter of 2019; however, output in non-IP-intensive industries was about 85% relative to the last quarter of 2019. Both IP-intensive and non-IP-intensive industries gradually recovered

3 See The White House (2021) for discussions of these policies.

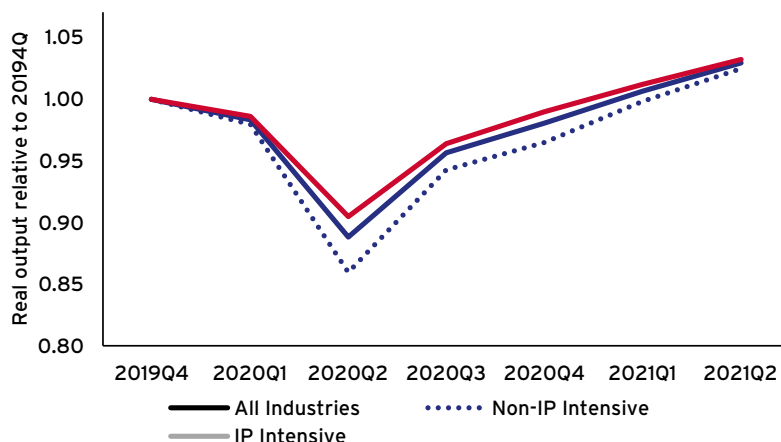
4 IP-intensive industries include the patent-intensive industries such as chemicals, electrical and electronic equipment, computers and communications equipment, medical devices, motor vehicles, ship, railroad, and aerospace building; copyright-intensive industries such as sound recordings, motion pictures, software publishing, newspapers, radio and television broadcasting, data processing; and trademark-intensive industries such as manufacturing and services (overlapping with copyright and patent industries). For more about US IP-intensive industries, see Economics and Statistics Administration and USPTO (2012).

5 The U.S. Centers for Disease Control and Prevention COVID Data Tracker contains extensive information on COVID-19 cases, deaths and more (<https://covid.cdc.gov/covid-data-tracker/#demographicsovertime>).

6 For a chronology of US business cycle events, see www.nber.org/research/business-cycle-dating.

their output of goods and services. By early 2021, the IP-intensive industries reached the same level of real output as in the last quarter of 2019. The non-IP-intensive industries would do so by the end of the first quarter of 2021.

FIGURE 1 US REAL GROSS OUTPUT



Note: Gross real output by industry. The gross output measure, seasonally adjusted at annual rates, is used, and converted into real 2012 dollars using its chain-type price index.

Source: Authors' calculations based on US Bureau of Economic Analysis Interactive Tables (www.bea.gov).

2 THE US INNOVATION EXPERIENCE, 2019-2021

2.1 Knowledge investments: R&D

From our snapshot look at real output, companies that innovate appear to stand apart from the general economy. Digging deeper into the different kinds of investments companies can undertake – namely, investments in R&D versus investments in physical capital – we observe further distinguishing features of innovation, particularly its relative stability over the business cycle.

Data on R&D and physical capital investments were obtained from 652 leading patenting companies in the US. These companies accounted for 83.3% of private sector R&D in 2019.⁷ Looking at this group of companies permits a comparison between intangible and tangible investment behaviour. Indeed, Table 1 shows a contrast in behaviour between physical capital investment and investments in R&D.⁸ First, it is apparent that the rate of physical capital investment is far more volatile than R&D. The rate of capital investment to sales fluctuated from -12.3% to 27.6%, while the rate of R&D to sales remained within the range of 7.3% to 8.5%.⁹

7 For this calculation we used Table 2 of the National Science Foundation (NSF) National Patterns of R&D Resources 2019-2020 Data Update, which reports that business R&D was \$443.6 billion (in real 2012 dollars) in 2019. The companies in our dataset accounts for over four-fifths of this amount.

8 The sources of data are in the appendix table.

9 The negative value of capital investment is attributable to gross investment being less than capital depreciation.

TABLE 1 INVESTMENT RATES: PHYSICAL CAPITAL INVESTMENT VERSUS R&D

	Rate of physical capital investment to sales	Rate of R&D investment to sales
2019:Q1	10.3	7.7
2019:Q2	1.9	7.4
2019:Q3	-1.6	7.3
2019:Q4	14.9	7.5
2020:Q1	27.6	8.1
2020:Q2	-12.3	8.5
2020:Q3	-4.5	7.8
2020:Q4	17.2	7.6
2021:Q1	-3.4	7.8
2021:Q2	3.3	7.6
2021:Q3	1.8	7.5
<i>Mean</i>	<i>5.0</i>	<i>7.7</i>
<i>Standard deviation</i>	<i>11.5</i>	<i>0.3</i>
<i>Coefficient of variation</i>	<i>2.3</i>	<i>0.04</i>

Notes: R&D is research and development expense; physical investment is the change in the stock of capital. The mean, standard deviation, and coefficient of variation are calculated over the period shown. The coefficient of variation equals the ratio the standard deviation to the mean, an indicator of the volatility (or spread) in the investment rates relative to the average investment rate. The bolded rows mark the period of the initial COVID-19 shock on business operations.

Source: S&P Capital IQ (www.capitaliq.com).

The relative volatility can be measured more formally with the coefficient of variation, which shows the ratio of the standard deviation (a common measure of the fluctuation or spread in values) to the mean.¹⁰ Whereas the coefficient of variation for the rate of physical capital investment is 2.3 – meaning that the amount of fluctuation in the rate of investment was more than twice that of the average physical capital investment rate – the coefficient of variation for the rate of R&D investment is only 0.04; that is, the amount of fluctuation was largely within 4% of the mean rate of R&D investment.

¹⁰ Dividing by the mean is especially useful since it makes the coefficient of variation scale-free. Standard deviations depend on the unit of measurement. It is best to view the spread in values relative to the mean to gain perspective.

The smoothness of R&D compared to physical capital investment can be partly explained by adjustment costs, particularly related to R&D employment.¹¹ However, it can also be explained by the opportunity to transfer R&D outcomes into valuable IP assets. These assets can be sold or licensed, making knowledge investments relatively more reversible (i.e. easier to recoup). Economic theory shows the value of postponing investments tends to increase as uncertainty rises, particularly for less reversible investments such as many forms of physical capital (e.g. buildings, and equipment).

A second notable aspect of the results is that investment in physical capital declined while that of R&D increased during the COVID-19 recession (see the second quarter of 2020 in Table 1). The rate of physical capital investment declined further in the third quarter of 2020 before rebounding in the fourth quarter of that year. The rate of R&D investment held steady as these fluctuations in physical capital investment occurred. By the third quarter of 2021, the rate of physical capital investment was still lower than at the start of the pandemic recession. The rise in R&D investment to sales in the second quarter of 2020 reflects investments in the health care, as medical devices and pharmaceutical companies responded to the spike in demand caused by COVID-19 and participated in emerging responses (e.g. Operation Warp Speed). It also reflects the COVID-driven shift in market demand from face-to-face transactions to digital commerce. Remote work and schooling increased the demand for digital tools for business and education.

2.2 Patenting

Turning to patents, we focus on the number of utility (or ‘invention’) patent applications filed at the US Patent and Trademark Office (USPTO) in three technological areas: digital, health, and all other ‘traditional’ technologies. Each area is defined by grouping patents using their technology classification assigned by the USPTO under the Cooperative Patent Classification (CPC) system.¹²

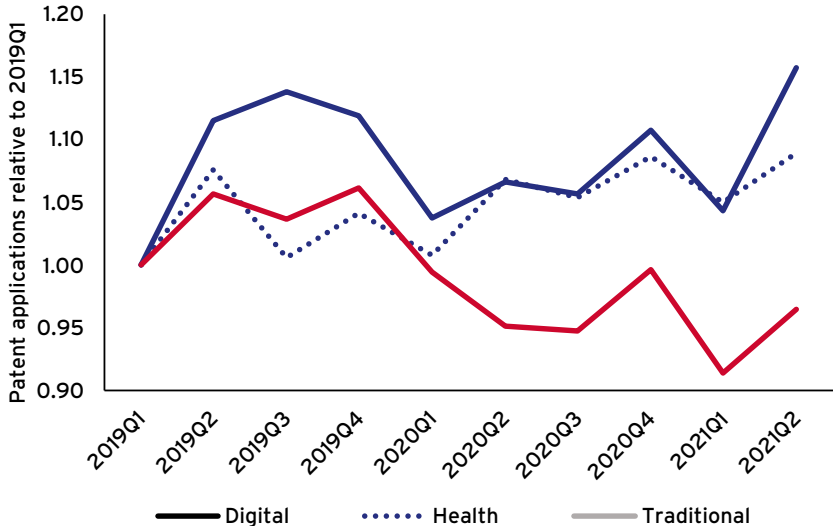
Figure 2 anchors the graphs of patent applications to the first quarter of 2019; that is, the applications are indexed to one (by dividing through by the number of applications filed in the first quarter of 2019). Thus, the graphs represent the growth of applications over time and indicate whether, at any given point, applications were higher or lower than they were in the first quarter of 2019. Consistent with the developments in R&D, the figure shows that patent applications in the digital and health areas by US-based inventors rose shortly after the COVID-19 recessionary shock. Patenting in these fields continued to stay robust during the summer of 2020 and grew faster by the fourth quarter of 2020, just before the vaccination rollout. We observe differences across technological fields. During the initial and later pandemic periods, the number of patent applications from

¹¹ Laying off and rehiring high-skilled R&D workers imposes substantial adjustment costs on companies.

¹² Applications are classified using the most recent, first-listed CPC symbol for the application's patent family. CPC symbols are mapped to WIPO technology classification using a concordance provided by WIPO and in some instances supplemented by our judgement. Information on the classification of the technology in the application as well as data on the applicant and assignee of the application are only available through the end of June 2021. This limitation does not apply to the trademark application data.

the traditional technologies remained below what they were in the first quarter of 2020. In contrast, patenting in digital and health technologies are above pre-pandemic levels in the second quarter of 2021 and greater than in the first quarter of 2020.

FIGURE 2 PATENT APPLICATIONS FROM US-BASED INVENTORS BY INDUSTRY GROUP



Notes: Utility (invention) patent applications filed by receipt date, the date which the application is initially submitted to USPTO. The first-listed CPC symbol for each application is used to classify each application. Origin counts are fractionalised based on each inventor's country code in their mailing address.
 Source: Author's calculations based on data from USPTO's Patent Location and Monitoring system (PALM).

Our findings are consistent with Davis et al. (2021), who compare the effects of the COVID-19 recession on employment between STEM (science, technology, engineering, and mathematics) occupations and non-STEM ones. While their focus is on education, interesting parallels can be drawn between STEM education and patent activity, both being important for innovative knowledge-intensive companies. Their study finds STEM employment was more resilient against the effects of the recession. Accounting for various factors that could underlie employment differences between the two broad occupation categories – such as demographics, educational attainment, remote work feasibility, geographic location, among others – the authors find ‘STEM knowledge on the job’ to be the critical factor.¹³ Possession of such knowledge enables workers to find and retain jobs that require fewer routine or manual tasks, making them less vulnerable to trends in automation and job retrenchment during downturns.

¹³ The study uses the O*NET database to derive a measure of the importance of STEM knowledge by occupation. The six STEM knowledge areas considered are: computers, engineering, mathematics, physics, chemistry, and biology. Survey questions would ask respondents how important each of these kinds of knowledge is to the performance of a job.

Davis et al. (2021) also present data on R&D employment, expenditures, and patent applications. The study shows that employment in R&D-intensive industries fell less than overall STEM employment during the first quarter of 2020, followed by a quick recovery.¹⁴ Likewise, R&D spending and patenting also fell during that quarter or shortly thereafter, and then rebounded quickly. The study infers that COVID-19 had a mild effect on US inventive activity and the authors conclude that this was “possibly enabled by STEM employment resiliency” (Davis et al. 2021). Their findings are consistent with our analyses of R&D, patenting, and the characteristics that help make US innovation more resilient, particularly the role of IP assets.

2.3 Trademark filings

Trademarks are commonly associated with the creation and protection of brands, but researchers have also highlighted the usefulness of trademark data as an indicator of innovation and entrepreneurship (Flikkema et al. 2014, Dinlersoz et al. 2018, Castaldi et al. 2020, deGrazia et al. 2020). Compared to patents, trademarks provide a broader signal of innovation by covering more segments of the US economy. For instance, out of 210 US industries considered in a recent USPTO report, 110 are trademark-intensive while only 70 are utility patent-intensive (Toole et al. 2022). Moreover, companies can design and apply for trademarks relatively quickly in response to product- and market-specific business conditions, making trademarks a potential leading indicator of economic outcomes (deGrazia et al. 2020).

The association between trademarks and entrepreneurship largely reflects the use of trademarks by new businesses. Dinlersoz et al. (2018) find that the probability of applying for a trademark for the first time, across all industries, was highest for startups and younger companies less than five years old. During the dot.com boom of the late 1990s, applications for trademarks surged (Graham et al. 2013). This coincided with a surge in new business formations, particularly in the ‘High Tech’ area, according to Goldschlag and Miranda (2016). Trademarks declined sharply in 2007 at the onset of the Great Recession, as did new employer businesses (Dinlersoz et al. 2018, 2021).

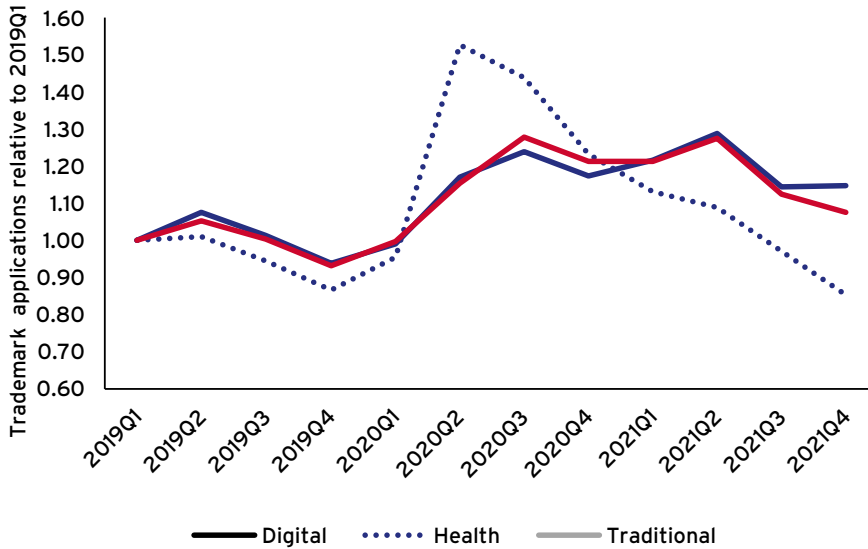
Figure 3 presents trademark applications in goods and services categories that correspond to our three areas of technology: health, digital, and traditional. The filing trends in the US were relatively stable just prior to the pandemic in 2019. They started to edge up at the earliest onset of the pandemic, in the first quarter of 2020. The early trademark response probably reflects decisions by US businesses to pivot production and modes of product and service delivery in response to the emerging pandemic.¹⁵ However, as the pandemic progressed, trademark application filings rose significantly above trend, before

14 The industries examined are computer systems design services, pharmaceutical and medicine manufacturing, software publishers, computer and electronics manufacturing, and scientific R&D services.

15 According to detailed information collected by the US Census Bureau’s Small Business Pulse Survey (SBPS) in April 2020, 6.8% of US businesses shifted production of their goods and services, and 15.2% changed the mode of delivery (Buffington et al. 2021b).

gradually returning to their pre-pandemic levels in 2021. This closely tracks new business registrations documented in Haltiwanger (2021). As the new business registrations spiked in the ‘Professional, Scientific and Technical Services’ industries in 2020, so did the number of trademark filings in digital and health-related products and services. Meanwhile, strong growth in new business registrations in the ‘Retail, Construction, and Trucking and Warehousing’ industries boosted filings in our traditional grouping of trademarks.

FIGURE 3 TRADEMARK APPLICATIONS FROM US-BASED ENTITIES BY INDUSTRY GROUP



Notes: Application classes filed by filing date. Application counts are based on the number of primary NICE classes in the application on the date of filing. The origin is determined by the address information provided by in assignment/ownership records.

Source: Authors' calculations based on USPTO Trademark Reporting and Monitoring System (TRAM).

3 UNCERTAINTY AND THE IMPORTANCE OF SECONDARY MARKETS

Thus far, the three takeaway points emphasised at the outset have been addressed: innovative industries and companies have weathered the COVID-19 pandemic better; innovation investments, such as R&D, behave differently than tangible, physical capital investments; but even among innovative companies, heterogeneity exists in patenting and trademarking trends between digital, health, and other technologies. This section addresses why those technologies, particularly in the digital area, may be relatively more resilient to a cyclical shock.

One clear impact of the COVID-19 pandemic was to increase uncertainty about the economy (Altig et al. 2020, Baker et al. 2020). To understand how greater uncertainty influences choices, economists often use a ‘real options’ framework (Dixit and Pindyck 1994, Abel et al. 1996). This framework has been developed primarily in the context of

physical capital investments – expenditures today to construct production facilities or buy equipment that will yield a stream of returns in the future – but it can also be used to interpret a variety of decisions such as investments in knowledge, decisions to open and close real assets such as coal mines, or even decisions to hire or fire personnel. Dixit and Pindyck (1994) highlight three critical characteristics for using a real option framework: (1) sunk costs, where some fraction of the expenditure cannot be recouped at a later date (also called ‘irreversibility’); (2) the presence of uncertainty about the economic environment; and (3) the ability to delay the investment opportunity.

In the context of COVID-19, a real options perspective suggests that greater uncertainty increases the value of waiting so that decision makers are likely to hold back on investments. This could help explain why the ratio of physical capital investment to sales falls in Table 1, even in high-tech industries. However, uncertainty also increases the ‘zone of inactivity’, meaning ongoing projects may not be abandoned as quickly. For knowledge-related investments such as R&D, projects are likely to be maintained, at least for a while. But there is another very important influence on decision makers in industries that use IP, namely, the secondary market for patents, trademarks, and copyrights. IP acts as a store of value, and opportunities to sell or license IP make investments related to IP more reversible. In other words, these investments can be recouped, at least to some degree, and this offsets the sunk cost condition. Park et al. (2021) explore the influence of exposure to secondary markets for patents and find that greater exposure helps mitigate the effects of economic uncertainty on a company’s decision to file for a patent.

To illustrate this point, Table 2 shows the distribution of IP trading deals for a small but informative sample obtained from a website hosted by IAM Market. The deals include licenses, cross-licenses, and sales, and are broken out by industry. Each deal often involves one or more related patents. Most deals (44%) took place in the digital area (electronic and electrical equipment, software and computer services, network communications, and computing and information technology hardware). This is perhaps not too surprising, as these are areas where innovation can be sequential, cumulative, fragmentable, and therefore constitute tradeable inputs into multiple products.

The health area (medical devices, biotechnology, and pharmaceuticals) appears in 13% of all deals. Here, the secondary market share is relatively smaller than that of the digital technologies, but is almost twice that of the third leading industry group involved in IP transactions, namely, the automobile and parts sector.

TABLE 2 SECONDARY MARKET DEALS IN PATENT RIGHTS: INDUSTRY REPRESENTATION

Industry group	Counts	Share (%)
Digital technologies	393	44.0
Health technologies	116	13.0
Automotive and parts	65	7.3
Telecommunications	46	5.2
Manufacturing	46	5.2
Media & entertainment	31	3.5
Electricity	27	3.0
Transport	27	3.0
Security & defence	26	2.9
Household goods	22	2.5
Chemicals	13	1.5
Food, tobacco, & beverages	13	1.5
Oil & gas	10	1.1
Finance	10	1.1
Retail & distribution	10	1.1
Construction & building materials	7	0.8
Test and measurement	7	0.8
Support services	6	0.7
Steel & other metals	5	0.6
Utilities & mining	5	0.6
Leisure & hospitality	4	0.4
Printing	3	0.3
Education	1	0.1
TOTAL	893	100.0

Note: Counts of transactions for all technology categories. Each transaction, or deal, consists of one or more patents.

Source: IAM Market (<https://portal.iam-market.com/browse-iptech>, accessed 21 March 2022).

The sample of secondary market deals is therefore quite skewed around a few industry groups, which are the industries demonstrated to be relatively least susceptible to the COVID-19 shock or to have rebounded swiftly from it. The ability to sell IP assets and recoup investments, even if the companies do not actually engage in sales, provides an insurance-like protective effect and thereby helps maintain the momentum to invest in innovative efforts.

4 CONCLUSIONS

Our chapter illustrates the resiliency of innovation despite the difficulties and challenges of the COVID-19 era. While the overall economy experienced losses in output, with consumers and workers contending with the upheavals, IP-intensive industries bore smaller initial losses and rebounded more quickly. R&D exhibited minimal cyclical fluctuation, while investment in physical capital was much more volatile in response to the economic shocks. Patenting and trademarking activities also remained relatively robust, if not bolstered, amid the economic shocks and uncertainties in the economy owing to persistence in the spread and impact of the virus.

Furthermore, while innovation was resilient overall, there exists heterogeneity across technological fields and areas of innovative activity. For example, patenting in the area of digital technologies, with its access to secondary markets and opportunities provided by the transition to virtual and remote work environments, and patenting in the health area, with its call to respond to the public health crisis, fared relatively well compared to the traditional technology areas. Trademarking activities in the digital and traditional goods and services remained strong given the new business creations that found opportunity during the pandemic.

Going forward, it is useful to reflect upon the factors that give innovation its resilience – for example, public policy support, institutions, the educational system, the market environment, organizational competence, and coordination among the different organisations and institutions, to name a few. This chapter has highlighted a few of the important links, focusing primarily on innovative US companies and industries – those that invest in knowledge-generating activities, use high-skilled labour such as workers with STEM degrees, and take advantage of the intellectual property system. STEM education provides the training for inventors and creators, IP institutions (intellectual property offices, international organisations) provide the means for inventors and creators to obtain property rights protection, and secondary markets provide a venue for trading IP rights. Innovation requires resources – financial, human, and technological – as well as infrastructural and institutional reinforcements. The important role that innovation has played in responding to the COVID-19 crises should make us reflect more deeply on the various components that sustain the overall innovation system. If these components and other support mechanisms had not functioned at a high level, the US innovation experience may not have shown the resiliency we observed during the pandemic.

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